ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Low oxygen requires fast breathing.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Spinal cord reflex centre.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Copying errors produce small variations, which explains the statement.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Fusion with X or Y sperm decides the sex.
5	**(c)** 6	$C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O$.
6	**(c)** diabetes	Insulin.
7	**(d)** variation, and those from the cutting are identical to the parent	Sexual gives variation; vegetative does not.
8	**(b)** 50	1/4 of 200 = 50 (Rryy).
9	**(a)** uses carbon dioxide and releases oxygen	Photosynthesis replenishes oxygen.
10	**(a)** adrenaline	Fight or flight.
11	**(b)** undifferentiated cells to produce missing parts	Specialised dividing cells are essential.
12	**(b)** 1 : 1 : 1 : 1	Four gamete types x one.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Turgid guard cells bulge apart and open the pore; shrunken ones close it.
14	**(a)** Both A and R are true and R is the correct explanation of A.	The short reflex arc through the spinal cord saves time, which makes the response quick.
15	**(a)** Both A and R are true and R is the correct explanation of A.	With a single parent and no gamete fusion there is no mixing of DNA, so the offspring are almost identical clones.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Volume per minute = 72 x 70 (1); = 5040 mL (5.04 L) (1).	1+1
17	Motor neuron or effector (1); Sensory path is intact (1).	1+1
18	Regeneration (1); It involves growth of a whole body from a body part, not an outgrowth (1).	1+1
19	TT fraction = 1/4 (1); 60 plants (1).	1+1
20	Bile neutralises acid from the stomach (1); Pancreatic enzymes work in alkaline medium (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Light is needed for photosynthesis (1); Starch is formed in light only (1); Covered region shows no starch (1).	1+1+1
22	Tip makes auxin (1); Auxin goes to the shaded side (1); No tip: no auxin, no bending (1).	1+1+1
23	Divisions in 2 hours = 8 (1); Number = 2^8 = 256 (1); Binary fission (asexual) (1).	1+1+1
24	Cross with green wrinkled (yyrr) (1); All yellow round offspring means YYRR (1); Other phenotypes in offspring mean the plant is heterozygous (1).	1+1+1
	OR: Heart works continuously (1); Needs oxygen and glucose (1); Blockage leads to heart attack (1).	1+1+1
25	Motor nerve carries signal to muscle (1); Effector is the muscle (1); Sensory part not needed (1).	1+1+1
	OR: Plants have meristematic cells for new growth (1); Humans have complex organ systems (1); Human body parts cannot make a whole body (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Salivary amylase (1). (ii) Hydrochloric acid (1). (iii) Small intestine (1). (iv) Bile emulsifies fats, breaking large globules into smaller ones for the action of lipase (1); the pancreatic enzymes act only in an alkaline medium, so the acidic food from the stomach is made alkaline by bile (1).	1+1+1+2
27	Asexual offspring are nearly identical clones (1); DNA copying errors give small variations only (1); Sexual reproduction mixes DNA of two parents (1); Variations help some survive changes in the environment (1); Example: cross-pollination in plants gives varied seeds (1).	1+1+1+1+1
	OR: F1: TtRr, tall round (1); F2 ratio 9 : 3 : 3 : 1 (1); Tall round 900, tall wrinkled 300 (1); Dwarf round 300, dwarf wrinkled 100 (1); Dwarf round genotypes ttRR and ttRr (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Germ-cell DNA is unchanged, so acquired traits are not passed on.
2	**(c)** 1	$6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$.
3	**(c)** reflex action is faster than brain sensation	Information reaches the brain later.
4	**(d)** spores need moisture to germinate	Water is needed to activate spores.
5	**(c)** 1/4	Each birth is independent.
6	**(b)** chlorophyll is needed for photosynthesis	Chlorophyll.
7	**(b)** iodine	Goitre.
8	**(a)** shortage of nutrients and oxygen	The placenta is the exchange site.
9	**(b)** new combinations of independently inherited genes	Independent assortment gives new combinations.
10	**(c)** A is true but R is false.	Bile has no enzymes; it emulsifies fats.
11	**(c)** A is true but R is false.	A true; R false, roots grow downward.
12	**(a)** Both A and R are true and R is the correct explanation of A.	The scrotum keeps the testes cooler.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Every F1 plant is Tt; the dominant allele T masks t, so R explains A.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; arteries carry blood under high pressure (hence thick walls) while valves are a separate feature of veins.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; they are different hormones, so R does not explain A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Feedback mechanism (1); Level depends on need (1).	1+1
17	Unisexual flower (1); Pollen must be moved from the male flower to the female flower (1).	1+1
18	T is dominant over t (1); The tall allele masks the dwarf allele (1).	1+1
19	Low haemoglobin (1); Less oxygen to cells, less energy (1).	1+1
20	Adrenaline released (1); Prepares the body for action (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Sperm 23 (22 autosomes) (1); Egg 23 (22 autosomes); zygote 46 (44 autosomes) (1); Liver cell 46 (44 autosomes) (1).	1+1+1
22	Salivary amylase breaks starch to maltose (1); Stomach acid stops amylase (1); Pancreatic amylase in small intestine gives glucose (1).	1+1+1
23	Adrenaline raises sugar for energy (1); Insulin lowers sugar (1); Opposing action (1).	1+1+1
24	Embryo would not implant (1); Pregnancy would not be established (1); Embryo could not be nourished (1).	1+1+1
	OR: A woman has two X chromosomes (XX) (1); Each egg receives only one of them (1); So each egg has one X and 22 autosomes (1).	1+1+1
25	Atmospheric oxygen is lower (1); Faster breathing to get more oxygen (1); More RBC for transport (1).	1+1+1
	OR: Eyes send signals to brain (1); Brain stimulates adrenal glands (1); Adrenaline prepares for flight (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Reflex action (1). (ii) Motor neuron (1). (iii) Spinal cord (1). (iv) A reflex is quick, involuntary and does not need thinking (1); writing is voluntary, controlled by the cerebrum and slower (1).	1+1+1+2
27	Egg is released from the ovary into the oviduct (1); Sperm fuses with the egg in the oviduct (1); Zygote divides and moves towards the uterus (1); Embryo is implanted in the thick lining (1); Placenta forms from embryonic and uterine tissue (1).	1+1+1+1+1
	OR: First two plants: dwarf offspring means each has t (1); 1/4 dwarf shows both are Tt (1); Third plant crossed with tt gives all tall (1); So it must be TT (1); If it were Tt, half would be dwarf (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 8	Three divisions give $2^3 = 8$.
2	**(a)** Tt	1 : 1 ratio.
3	**(a)** both plants	No light or no CO2 means no photosynthesis.
4	**(a)** gigantism	Excess growth hormone gives excessive growth.
5	**(a)** The egg cannot meet the sperm	Fertilisation occurs in the oviduct.
6	**(b)** environment affects expression of the trait	Phenotype depends on environment.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Transpiration pull.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Prevents goitre.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Equal gametes T and t give TT : Tt : tt = 1 : 2 : 1.
11	**(c)** 5040	70 x 72.
12	**(a)** not feel a stimulus	Information does not reach CNS.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; they are different modes of asexual reproduction, and R does not explain A.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the second statement concerns the sex chromosome carried by sperm and does not explain the equal contribution.
15	**(c)** A is true but R is false.	Anaerobic breakdown is incomplete and releases much less energy than aerobic respiration; R is false.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Pollen grain (1); Formed in the anther (1).	1+1
17	Ff x ff gives 1 : 1 (1); 2 children (1).	1+1
18	Reabsorbed 178.5 L (1); Percentage = 99.2% (1).	1+1
19	Roots grow into the soil (1); Absorb water and minerals (1).	1+1
20	Number of divisions = 6 (1); $2^6 = 64$ Amoebae (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Ovaries release the egg and secrete oestrogen (1); Oestrogen controls thickening of the uterine lining (1); Without it the lining is not formed or shed (1).	1+1+1
22	Each pregnancy is independent (1); X and Y sperm are equally likely (1); Probability = 1/2 (1).	1+1+1
23	Tubule reabsorption is faulty (1); Normally all glucose is reabsorbed (1); Diabetes may be a cause (1).	1+1+1
24	Chemicals from ovule attract (1); Chemotropism (1); Growth of pollen tube (1).	1+1+1
	OR: Pollen tube must grow a long distance (1); It grows through the style tissue (1); It delivers male gametes to the ovule (1).	1+1+1
25	Green beetles are less easily seen (1); They survive and reproduce more (1); The proportion of green beetles increases (1).	1+1+1
	OR: Soil solution is more dilute (1); Water moves into root hair cell (1); Moves into xylem (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Stamen (anther produces pollen) (1). (ii) Pollination (1). (iii) Ovary (1). (iv) Self-pollination: pollen from the anther reaches the stigma of the same flower (or another flower of the same plant); cross-pollination: pollen reaches the stigma of a flower of a different plant (1); agent: wind, water, insects or animals (any one) (1).	1+1+1+2
27	Moist and warm: rapid growth of mould (1); Dry: little or no growth (1); Moist and cold: slow growth (1); Spores germinate only with moisture and warmth (1); Thick-walled spores wait for suitable conditions (1).	1+1+1+1+1
	OR: Tt x tt gives 1 tall : 1 dwarf (1); Rr x Rr gives 3 round : 1 wrinkled (1); Combine the independent ratios (1); Tall round 3, tall wrinkled 1, dwarf round 3, dwarf wrinkled 1 (1); Ratio 3 : 1 : 3 : 1 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND